

1 Satellite tile

- 0.5 m per pixel, RGB
- 1024 x 1024 tile
- Training: random
- 256 x 256 crops, flips

RGB

2 MobileViT v2 network

- 1.60 M parameters
- Separable attention $O(N \cdot d)$
- Strip conv $1 \times 3 \rightarrow 3 \times 1$
- Channel shift, no weights
- Attention gated skips

P

3 Postprocessing

- Four flip TTA
- Hysteresis $T_h = 0.35$, $T_l = 0.12$
- Closing 5×5
- Gap bridging $d \leq 220$ px
- End point angle $\theta \leq 65^\circ$

mask

4 Graph, deployment

- Skeleton to $G = (V, E)$
- Betweenness $C_B(v)$
- Resilience index R
- ONNX Runtime, 6.9 MB
- 0.91 s per tile on CPU

5 Ground truth

- DeepGlobe road mask Y
- Binary, 0.5 m per pixel

Y

6 Soft skeleton

- Min and max pooling
- $K = 10$ iterations
- Gives $S(P)$ and $S(Y)$

$S(\cdot)$

7 Composite loss and model selection

- $\mathcal{L} = \alpha(e) \mathcal{L}_{BCE} + \beta \mathcal{L}_{Dice} + (1 - \alpha - \beta) \mathcal{L}_{clDice}$
- α decays from 0.50 to 0.15 over 40 epochs, $\beta = 0.35$
- Reject epochs that mark more than 20% of pixels as road

gradient